Chapter Four: Measures of central tendency

Definition:

An average is a single value that is considered as the most representative or typical value for a given set of data. Such a value is neither the smallest nor the largest value, but is a number whose value is somewhere in the middle or centre of the group. For this reason, an average is frequently referred to as a measure of central tendency shows the tendency of some central values around which data tends to cluster around.

Objectives of averaging:

There are two main objectives of the study of averages:

i) To get one single value that describes the characteristics of the entire data.

ii) To facilitate comparison.

Types of average:

The following are the important measures of central tendency which are generally used in business:

i) Arithmetic mean, ii) Median, iii) Mode, iv) Geometric mean, v) Harmonic mean.

Arithmetic mean:

The arithmetic mean, often simply referred to as mean, is the total of the values of a set of observations divided by their total no. of observations. Thus if X_1, X_2, X_3, X_4, ...,X_n represents the values of N items of observations, the arithmetic mean denoted by $\overline{X}$ is defined as:

$$\overline{X} = \frac{X_1 + X_2 + \dots\dots\dots\dots + X_n}{n}$$

$$= \frac{\sum_{i=1}^{n} X_i}{n}$$

If the subscripts are dropped, the formula becomes, $\overline{X} = \frac{\sum X}{n}$

For frequency distribution, it is given by

$\overline{X} = \frac{\sum fX}{n}$, where X is the mid value, f is the frequency of the class, n is the total frequency.

Example 01: The monthly income (in Tk.) of 10 persons working in a firm as follows:
1500, 1600, 1800, 1700, 1600, 1200, 1500, 2000, 1500, 1800

Find the average monthly income.

Solution:

Let income be denoted by X. $\sum X = 1500 + 1600 + \dots\dots\dots + 1800 = 16200$

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[1] Associate Professor of Economics of FSIS at IIUC and Research Associate of Islamic Economics Research Bureau of Bangladesh

$$\overline{X}=\frac{\sum X}{n}=\frac{16200}{10}=Tk.1620$$

Exercise 01:

Find A.M of the following observations:

20, 90, 80, 30, 65, 32

Exercise 02:

The production manager of Minar Press is determining the average time needed to photograph one printing plate. Using a stopwatch and observing the plate makers, he collects the following times (in seconds):

20.4, 20.0, 22.2, 23.8, 21.3, 25.1, 21.2, 22.9, 28.2, 24.3, 22.0, 24.7, 25.7, 24.9, 22.7, 24.4, 24.3, 23.6, 23.2, 21.0

An average per-plate time of less than 23.0 seconds indicates satisfactory productivity. Should the production manager be concerned?

Example 02:

1500 workers are working in an industrial establishment. Their age is classified as follows:

Age (years)	No. of workers
18-22	120
22-26	125
26-30	280
30-34	260
34-38	155
38-42	184
42-46	162
46-50	86
50-54	75
54-58	53

Find the mean age of workers.

Solution:

Age (years)	Mid value of age (X)	No. of workers(f)	fX
18-22	20	120	2400
22-26	24	125	3000
26-30	28	280	7840
30-34	32	260	8320
34-38	36	155	5580
38-42	40	184	7360
42-46	44	162	7128
46-50	48	86	4128
50-54	52	75	3900
54-58	56	53	2968

Total	….	n=1500	$\sum fX = 52624$

We have, A.M, $\bar{X} = \frac{\sum fX}{n} = \frac{52624}{1500} = 35.08$ years.

Mean age of 1500 workers will be 35.08 years.

Example 03:

1500 workers are working in an industrial establishment. Their age is classified as follows:

Age (years)	No. of workers
18-22	120
22-26	125
26-30	280
30-34	260
34-38	155
38-42	184
42-46	162
46-50	86
50-54	75
54-58	53

Find the mean age.

Solution:

Age(years)	Mid value (x)	No. of workers(f)	d= $\frac{x-A}{C}$ A=36, C=4	fd
18-22	20	120	-4	-480
22-26	24	125	-3	-375
26-30	28	280	-2	-560
30-34	32	260	-1	-260
34-38	36=A	155	0	0
38-42	40	184	1	184
42-46	44	162	2	324
46-50	48	86	3	258
50-54	52	75	4	300
54-58	56	53	5	265
		n=1500		$\sum fd = -344$

Indirect Method or Short-cut method: We have,

$$A, M., \cancel{X} = A + C \times \frac{\sum fd}{n} = 36 + 4 \times \frac{-344}{1500} = 36 - \frac{1376}{1500} = 36 - 0.9173 = 35.08 \text{ years.}$$

Median:

The median is defined as the middle most observation when the observations are arranged in order of magnitude.

For ungrouped data, when n is odd, the middle most observation i.e. the $(\frac{N+1}{2})th$ observation will be median in the series. Again when n is even, the median will be the arithmetic mean of $\frac{N}{2}th$ and $(\frac{N}{2}+1)th$ observations in the series. For example, the median of the observations 5,10,7,3,2 i.e. 10,7,5,3,2 is 5 and the median of 11,3,9,5,7,13 i.e. 13,11,9,7,5,3 is $\frac{9+7}{2} = 8$.

Exercise 1:

Find Median of the following observations:

i) 20, 90, 80, 30, 65, 32

ii) 20, 90, 80, 30, 65, 32, 35

Solution:

For grouped frequency distribution the median is given by

$Me = L + \frac{\frac{n}{2} - F}{f} \times c$, Where L is the lower limit of the median class (median class is that class which contains $\frac{n}{2}th$ observation of the distribution)

n is total number of observations

F is the preceding cumulative frequency to the median class

f is the frequency of the median class

c is the class interval of the median class

Example 01**:**

1500 workers are working in an industrial establishment. Their age is classified as follows:

Age (years)	No. of workers
18-22	120
22-26	125
26-30	280

30-34	260
34-38	155
38-42	184
42-46	162
46-50	86
50-54	75
54-58	53

Find the median age of the workers.

Solution:

Age (years)	No. of workers(f)	Cumulative frequency
18-22	120	120
22-26	125	245
26-30	280	525=F
30-34 (Median class)	260=f	785
34-38	155	940
38-42	184	1124
42-46	162	1286
46-50	86	1372
50-54	75	1447
54-58	53	1500
	n=1500	

We have, Median, $Me = L + \frac{\frac{n}{2} - F}{f} \times c$

Here, $\frac{n}{2} = \frac{1500}{2} = 750$, Median class =(30-34), L=30, f=260, F=525, c=4

$$Me = L + \frac{\frac{n}{2} - F}{f} \times c = 30 + \frac{750-525}{260} \times 4 = 30 + \frac{225}{260} \times 4$$

$$= 30 + \frac{900}{260} = 30 + 3.46 = 33.46$$

Hence, median age of 1500 workers will be 33.46 years.

Exercise 2**:**

The fuel consumption of 500 cars of a city are given below:

Fuel consumption (in liter)	0 – 100	100 – 200	200 – 300	300 – 400	400 – 500	500 – 600
No. of cars	50	125	200	75	40	10

Find median mathematically.

Mode:

Mode is the most typical or commonly observed value in a set of data. For example, if we take the values of six different observations, as 5,8,10,8,5,8 Mode will be 8 as it has occurred maximum number of times i.e. 3 times.

Exercise 1:

Find Mode of the following observations:

ii) 20, 90, 80, 30, 65, 32, 30

ii) 20, 90, 80, 30, 65, 32, 30, 80

Solution:

In case of grouped data or frequency distribution the following formula is used for calculating Mode:

$$Mode = L + \frac{f_1 - f_0}{f_1 - f_0 + f_1 - f_2} \times C$$

Where, L=Lower limit of the modal class (Modal class is that class for which the frequency is maximum)

f_1=frequency of the modal class

f_0=frequency of the class preceding of the modal class

f_2=frequency of the class succeeding of the modal class

Example 09**:**

The following data relate to the sales of 100 companies:

Sales(Tk. Lakhs)	No. of companies
Below 60	12
60-62	18
62-64	25=f_0
64-66 (Modal class)	30 =f_1
66-68	10=f_2
68-70	3
70-72	2
Total	n=100

Calculate the modal sales of the companies.

Solution:

We have,

$$Mode = L + \frac{f_1 - f_0}{f_1 - f_0 + f_1 - f_2} \times C$$

Since the maximum frequency 30 lies in the class 64-66, therefore 64-66 is the modal class.

Here, $L=64$, $f_1=30$, $f_0=25$, $f_2=10$, $C=2$

$$Mode = 64 + \frac{30-25}{30-25+30-10} \times 2 = 64 + \frac{5}{5+20} \times 2 = 64 + \frac{10}{25}$$

$$= 64 + 0.4 = 64.4$$

Hence, modal sales of the companies are Tk. 64.4 lakhs.

Exercise 1**:**

The fuel consumption of 500 cars of a city are given below:

Fuel Consumption (in liter)	0-100	100-200	200-300	300-400	400-500
No. of cars	80	125	200	75	20

Find modal fuel consumption mathematically.

Indirect method for calculating mode:

When mode is ill defined, its value may be ascertained by the following approximate formula based upon the relationship between mean, median and mode:

Mode=3 median – 2 mean

Exercise 14**:**

The life of a type of tires in a sample survey is given below:

Life (in km.)	No. of tires
5000-10000	200
10000-15000	500
15000-20000	500
20000-25000	300
25000-30000	125
Total	n=1625

Find modal no. of tires.

Solution:

We have, Median, $Me = L + \dfrac{\frac{n}{2} - F}{f} \times c$

and A.M, $\overline{X} = \dfrac{\sum fX}{n}$

Mode=3 median – 2 A.M.

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Merit:

i) It is not affected by extreme values.
ii) It can be obtained in open-end distribution.
iii) It can be easily used to describe qualitative phenomenon.

Demerit:

i) It is not clearly defined in case of bimodal or multimodal distribution.
ii) It is not based on all the observation.
iii) It is difficult for algebraic treatment.

Example 2 (Discrete frequency Distribution**):**
The frequency distribution of family members of 300 families of a village is shown in the following table:

Family member	1	2	3	4	5	6	7	8	Total
No. of families	28	35	45	65	52	32	26	17	n=300

Calculate mean, median and modal time.

Solution:

Family member (x)	1	2	3	4	5	6	7	8	Total
No. of families (f)	28	35	45	65	52	32	26	17	n=300
fx	28	70	135	260	260	192	182	136	1263
Cumulative frequency	28	63	108	173	225	257	283	300	--

A.M, $\bar{X} = \frac{\sum fX}{n} = \frac{1263}{300} = 4.21$

n/2=300/2=150, Median=4

Mode= 4

Geometric Mean:

Geometric mean is defined as the nth root of the product of n non-zero positive observations of a series.

Symbolically,

G.M.= $(X_1 \times X_2 \times X_3 \times .... \times X_n)^{\frac{1}{n}}$ Where X_1, X_2, X_3,.............. X_n, refer to n observations of a series. For example, the geometric mean of 2 and 8 will be G.M.= $(2 \times 8)^{\frac{1}{2}} = 16^{0.5} = 4$

For frequency distribution, G.M.=Anti-log $\frac{\sum f \log x}{n}$

Example 11: The frequency distribution of price (in Tk.) of 20 pens is given below:

Price	No. of pens
2-4	3
4-6	5
6-8	7
8-10	4
10-12	1
Total	n=20

Find Geometric mean of price of 20 pens.

Solution:

Price	No. of pens (f)	Mid value(x)	log x	f logx
2-4	3	3	0.48	1.44
4-6	5	5	0.70	3.50
6-8	7	7	0.85	5.95
8-10	4	9	0.95	3.80
10-12	1	11	1.04	1.04
Total	n=20			15.73

$$G.M. = \text{Anti-log}\frac{\sum f \log x}{n} = Anti-\log\frac{15.73}{20}$$

$$= \text{Anti-log } 0.787 = \text{Tk.}6.12$$

Therefore, Geometric mean of price of 20 pens will be Tk. 6.12

Application of Geometric Mean:

The geometric mean is used to find the average percent increase in sales, production, population or other economics or business series. It is theoretically considered to be the best average in the construction of index number.

Harmonic Mean:

Harmonic mean is defined as the reciprocal of the arithmetic mean of the reciprocal of the individual non-zero positive observations. Thus by definition

$$H.M. = \frac{n}{\sum \frac{1}{X}}$$

For example, the harmonic mean of 5,10,20,50,100 will be

$$H.M. = \frac{5}{\frac{1}{5}+\frac{1}{10}+\frac{1}{20}+\frac{1}{50}+\frac{1}{100}} = \frac{5}{0.2+0.1+0.05+0.02+0.01} = \frac{5}{3.8} = 1.32$$

For frequency distribution,

$$H.M. = \frac{n}{\sum \frac{f}{X}}$$

Example 12: The frequency distribution of price (in Tk.) of 20 shirts is given below:

Price	No. of shirts
200-400	3
400-600	5
600-800	7
800-1000	4
1000-1200	1
Total	n=20

Find Harmonic mean of price of 20 shirts.

Solution:

Price	No. of shirts (f)	Mid value(x)	f/x
200-400	3	300	0.01
400-600	5	500	0.01
600-800	7	700	0.01
800-1000	4	900	0.004
1000-1200	1	1100	0.0009
Total	n=20		0.0349

$$H.M. = \frac{n}{\Sigma \frac{f}{X}} = \frac{20}{0.0349} = 573.07$$

Therefore, Harmonic mean of price of 20 shirts will be Tk. 573.07

Application of Harmonic Mean:

The harmonic mean is a measure of central tendency for data expressed as rates, such as kms. per hour , tonnes per day, kms. per litre etc.